

Excellent 🙌 — these three tools are **core monitoring and troubleshooting utilities** in SQL Server, and every DBA or performance-focused developer should understand them deeply.

Let's go step by step with a **detailed and practical explanation** of each:

◆ 1. Extended Events (XEvents)

🧠 What It Is

Extended Events is a **lightweight performance monitoring framework** in SQL Server — designed to **collect and analyze event data** about what's happening inside SQL Server (queries, waits, deadlocks, errors, etc.).

It's the **modern replacement** for SQL Trace and Profiler — more efficient, flexible, and less resource-heavy.

⚙️ Key Features

- **Lightweight:** Minimal overhead on performance (better than Profiler).
 - **Highly customizable:** You can capture only what you need (events, filters, actions).
 - **Integrated with SSMS:** You can view live data or save to .xel file for later.
 - **Supports advanced diagnostics:** deadlocks, waits, IO, CPU, blocking, etc.
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🌱 Example: Capture Long-Running Queries

You can create an Extended Event session to monitor queries taking longer than 2 seconds.

```
CREATE EVENT SESSION LongRunningQueries
```

```
ON SERVER
```

```
ADD EVENT sqlserver.rpc_completed(
```

```
    WHERE duration > 2000000) -- duration in microseconds (2 sec)
```

```
ADD TARGET package0.event_file (SET filename = 'C:\XE\LongRunningQueries.xel');
```

```
GO
```

```
ALTER EVENT SESSION LongRunningQueries ON SERVER STATE = START;
```

➡ This logs all queries that take more than 2 seconds into a file.

You can view it in **SSMS → Management → Extended Events → Sessions**.

Typical Uses

- Detect **deadlocks, blocking, long queries, memory issues**.
 - Track **login/logout, index usage, failed queries**, etc.
 - Performance tuning with minimal server overhead.
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◆ 2. SQL Server Profiler

What It Is

SQL Server Profiler is an **older GUI-based tracing tool** that captures SQL Server events in real time — for debugging, auditing, or performance analysis.

It's based on **SQL Trace**, which is now deprecated but still widely used for quick analysis.

How It Works

You define a **Trace**, which listens for specific events like:

- SQL:BatchStarting / Completed (T-SQL commands)
- RPC:Completed (Stored procedure calls)
- Deadlock graph
- Errors and warnings

Then Profiler displays the trace live, showing:

- Login name, duration, CPU, reads/writes, SPID, etc.
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Example Usage

1. Open SSMS → Tools → **SQL Server Profiler**
 2. Connect to your server
 3. Choose a template (e.g., *TSQL_Duration*)
 4. Start trace → Run your queries
 5. Profiler displays which queries take time, CPU, reads, etc.
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Important Notes

- Profiler adds **overhead** if used on a production server.
 - Prefer **Extended Events** for live production tracing.
 - Good for **development**, debugging, or understanding workloads.
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Typical Uses

- Real-time monitoring of SQL activity.
 - Identifying slow stored procedures.
 - Capturing query workload for tuning.
 - Debugging application–database interactions.
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◆ **3. Activity Monitor**

What It Is

Activity Monitor is a **built-in graphical tool** in SSMS to quickly check SQL Server's real-time performance health.

It's like a **Task Manager for SQL Server**, showing live CPU, I/O, wait times, locks, and expensive queries.

How to Open

In SSMS:

Right-click Server → Activity Monitor

Key Sections

Section	Description
Overview	Shows CPU, I/O, Waits, Batch Requests/sec
Processes	Lists active sessions (SPIDs), login, database, wait type
Resource Waits	Displays top wait categories (e.g., PAGEIOLATCH, LCK_M_S)
Data File I/O	Reads/writes per database file
Recent Expensive Queries	Displays top queries by CPU, I/O, duration
Active Expensive Queries	Queries currently consuming most resources

Typical Uses

- Detecting blocking sessions (right-click → *Kill Process*).
- Viewing slow or expensive queries.
- Checking overall server performance quickly.
- Useful for quick health checks, especially for developers.

Summary Comparison

Tool	Purpose	Use Case	Performance Impact	Recommended For
Extended Events	Modern, lightweight monitoring	Deep analysis, deadlocks, performance tuning	Low	Production use
SQL Server Profiler	Legacy tracing tool	Debugging, dev tracing	High	Dev / Test

Tool	Purpose	Use Case	Performance Impact	Recommended For
Activity Monitor	GUI-based live dashboard	Quick health and blocking check	Medium	Admin monitoring

🔧 Practical Recommendation

Scenario	Tool
Find who is blocking others	Activity Monitor
Capture and analyze deadlocks or waits	Extended Events
Debug stored procedure from app in dev	Profiler
Analyze performance issues over time	Extended Events + Query Store